Step 1: Installing python

- We need version 3.14+ (python does not maintain strict compatibility between versions)

MS Windows:

- Download the **python install manager** from python.org and follow the instructions
- If you had previously installed python using the MS Windows Store packages, I may need to help you clean those up to get back to a sane system state

MacOS:

- Follow the instructions on python.org

Linux:

- Follow the usual process for your distribution. On ubuntu: sudo apt install python3-pip python3-venv

TEST: open a command/shell prompt, type in python<enter>, check version, exit by typing quit()<enter>

If you have previously used anaconda, we may be in for interesting times ;)

Step 2: Installing the EECI course codebase

- Check out the repository https://github.com/JochenTrumpf/EECI2026.git
- [optional] rename the folder to something you can remember
- Check that the EECI.code-workspace file is at the top level inside your new folder
- We will be using a pre-alpha version of the robosimpy library for control-oriented robot simulation

- Open a command/shell prompt in your new folder
- Create a python virtual environment by calling python –m venv .venv
- Activate the new virtual environment:
  - .\.venv\Scripts\Activate.ps1 (MS PowerShell)
  - .\.venv\Scripts\Activate.bat (MS Command)
  - source .venv/bin/activate (Ubuntu bash)

- Install necessary python packages
  - pip install jupyter numpy scipy tqdm plotly pandas nbformat dash matplotlib websockets
- If the new pip version release notice pops up, do what it tells you to do (optional)

Step3: Installing MS Visual Studio Code

- This is optional if you have a different preferred way to work with python scripts and jupyter notebooks

- Follow the installation instructions on https://code.visualstudio.com/download
- Launch vscode and install the **official Microsoft** Jupyter, Python and Python debugger extensions
- If vscode does not recognise your previously installed python version, **DO NOT PROCEED** with whatever it tells you to do. Ask me for help.

- Close **all** vscode instances

- Open the EECI.code-workspace file, this should start vscode in Workspace mode
- Answer “Yes, I trust the authors …” when prompted
- Choose View -> Command Palette -> Python: Select Interpreter

- Make sure to select the newly created venv environment under .venv/… (Workspace)
- This is the most critical step to make sure that notebooks and the debugger work properly

- Close **all** vscode instances

Step 4: Testing your installation

- Open vscode via the EECI.code-workspace file (or directly if that was the last thing you worked on)
- Select View -> Terminal and **WAIT** until the venv environment is automatically activated
- Check with me if this has not happened after a few seconds

- In the file list on the left, locate the file test.ipynb and click it
- This will open a test jupyter notebook in vscode

- In the top right corner, choose “Select Kernel”, choose the previously created venv environment

- You can now execute the jupyter notebook cell by cell in order (using the little play icon “Execute Cell” that pops up if you move your cursor over the cell)
- Check with me if something doesn’t work

- You are all set ☺